AI-BASED EARLY MENTAL HEALTH BREAKDOWN DETECTION FROM SPEECH PATTERNS

MileStone-5 Report
Presented By : Group -6

TEAM MEMBERS

Om Aryan (21f3002286)

Pankaj Mohan Sahu (21f2001203)

Drashti Shah (22f2001483)

Mahi Mudgal (21f3002602)

G Hamsini (22f3000767)

PIPELINE & MODELS RECAP

Two Pipelines:

- Pipeline 1 (Baseline): Raw Audio → MFCC/HRV Features → RobustScaler → XGBoost / SVM / MLP
- Pipeline 2 (Proposed): Raw Audio → Whisper Encoder (512-dim, frozen) → Fusion MLP

Four Model Families:

- XGBoost · SVM (RBF) · EmotionMLP · StressMLP

Dataset	Task	Test size	Key challenge
Ravdess	8-class emotion	300 files	Unseen speakers
DAIC-WOZ	Depression binary + PHQ-8	47 sessions	2.57:1 class imbalance
MODMA	MDD vs Healthy	8 subjects	Extremely small N
Swell-HRV	3-class stress	31k windows	Naturalistic stressor

Preprocessing & Environment

Preprocessing (no data leakage — scaler fitted on train only):

- MFCC: 338 features per file · Whisper: 512-dim per 30-sec chunk
- HRV: RMSSD, SDNN, LF/HF ratio · RobustScaler + outlier clipping

Environment:

- Platform: Kaggle Notebooks
- GPU: NVIDIA Tesla T4 — Whisper extraction & MLP training
- CPU: XGBoost & SVM
- Libraries: PyTorch 2.0 · Scikit-learn 1.3 · XGBoost 1.7 · Librosa

Metrics & Justification

Task	Metric	Reason
All Classification	Macro-F1	Equal weight per class : prevents majority-class bias
DAIC-WOZ	+ Sensitivity	Missing depression = clinical risk
DAIC-WOZ	AUPRC	More informative than AUC-ROC under 16% positive class
PHQ-* Regression	RMSE	Penalizes large errors — critical near the score-10 threshold

Note :Accuracy alone is misleading: a majority-class predictor gets 72% accuracy on DAIC-WOZ while detecting zero depressed patients

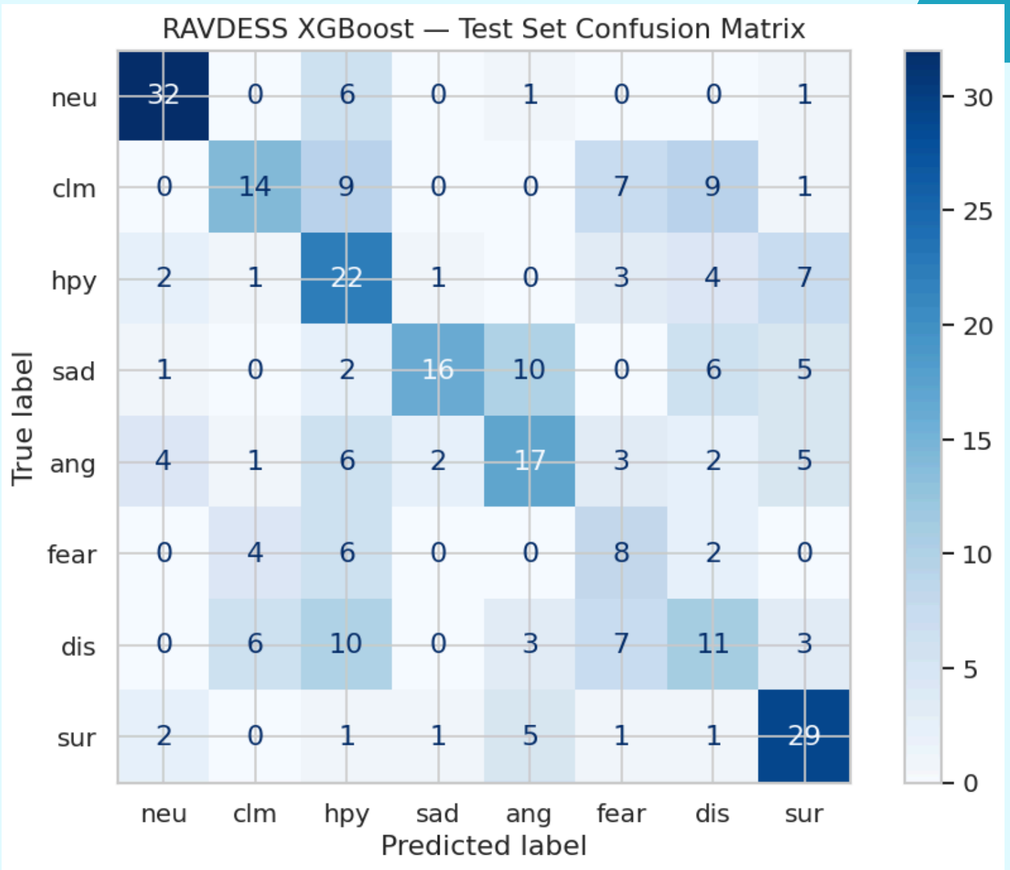
BEST RESULTS OF ALL TASKS

Dateset	Best Model	Accuracy	Macro-F1
RAVDESS	Whisper + XGBoost	0.972	0.974
DAIC-WOZ	Fusion MLP	0.674	0.607
MODMA	SVM(C=100, RBF)	0.875	0.873
SWELL-HRV	XGBoost	0.508	0.508

PHQ-8 BEST RMSE: 6.515 · BEST DEPRESSION SENSITIVITY: 0.857

RAVDESS Results

Model	Test Macro-F1
XGBoost (MFCC, best config)	0.486
EmotionMLP (AdamW)	0.33
SVM (RBF)	0.28
Whisper + XGBoost	**0.974**



WHISPER IMPROVES F1 BY +48.8 PERCENTAGE POINTS OVER THE BEST MFCC MODEL

DAIC-WOZ & MODMA RESULTS

DAIC-WOZ:

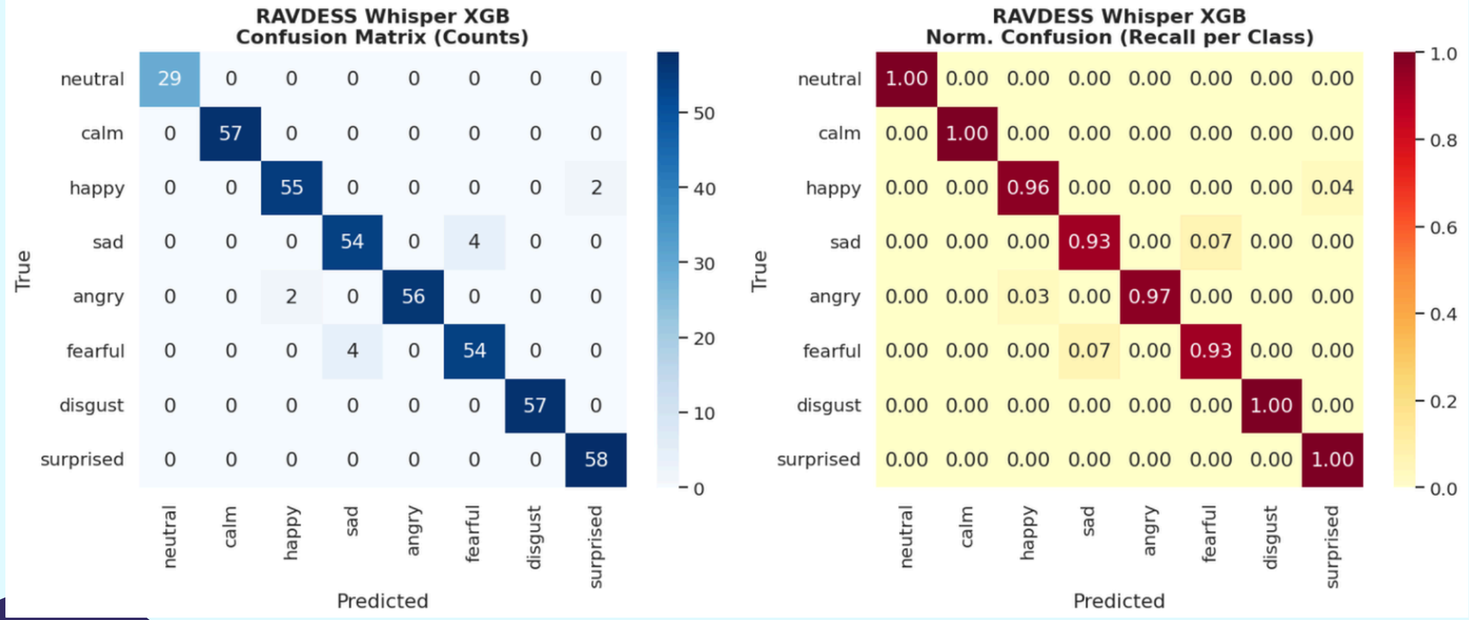
Model	Val F1	Test F1
Acoustic XGB only	0.5732	0.506
3-braind fusion	0.449	–
Fusion MLP	_`	0.607

MODMA:

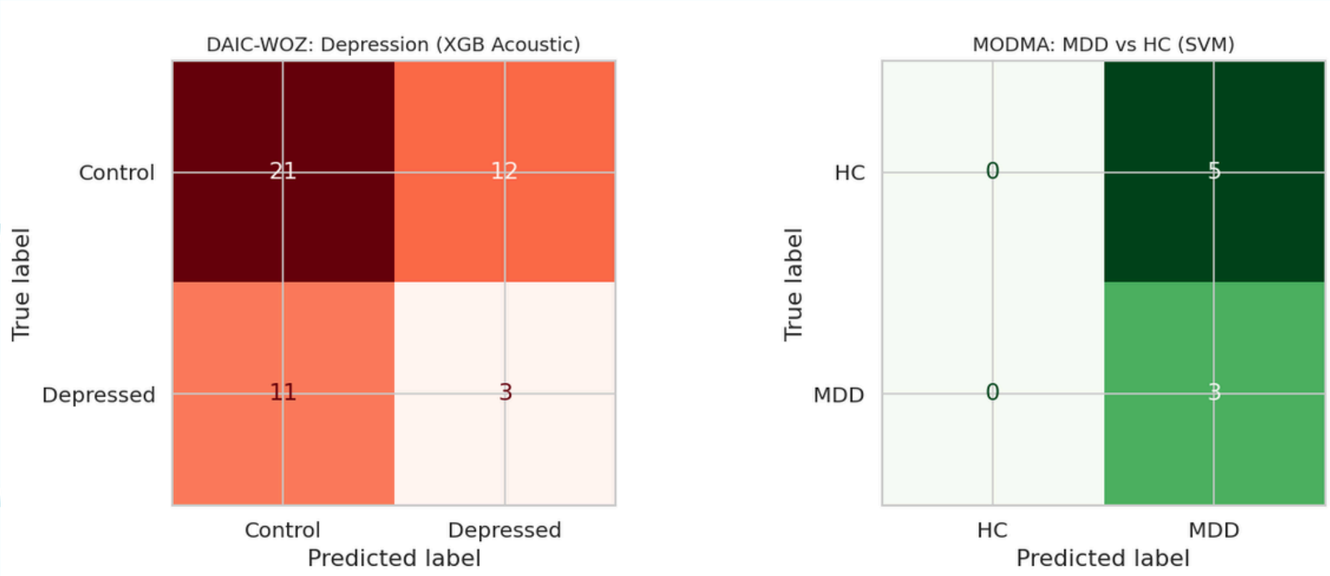
Model	Test F1
SVM (Classical Features)	0.873
Whisper+SVM	0.750

Whisper hurts MODMA — domain mismatch
(English ASR vs. Mandarin clinical audio)

CONFUSION MATRICES

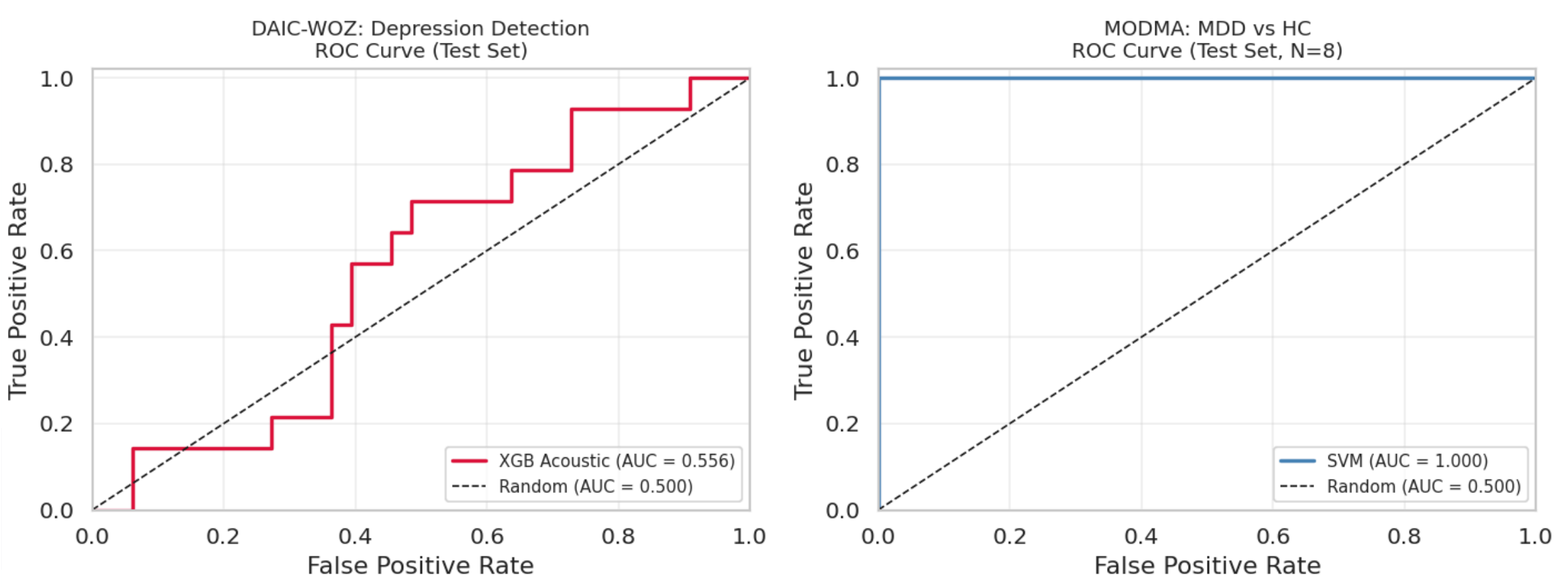


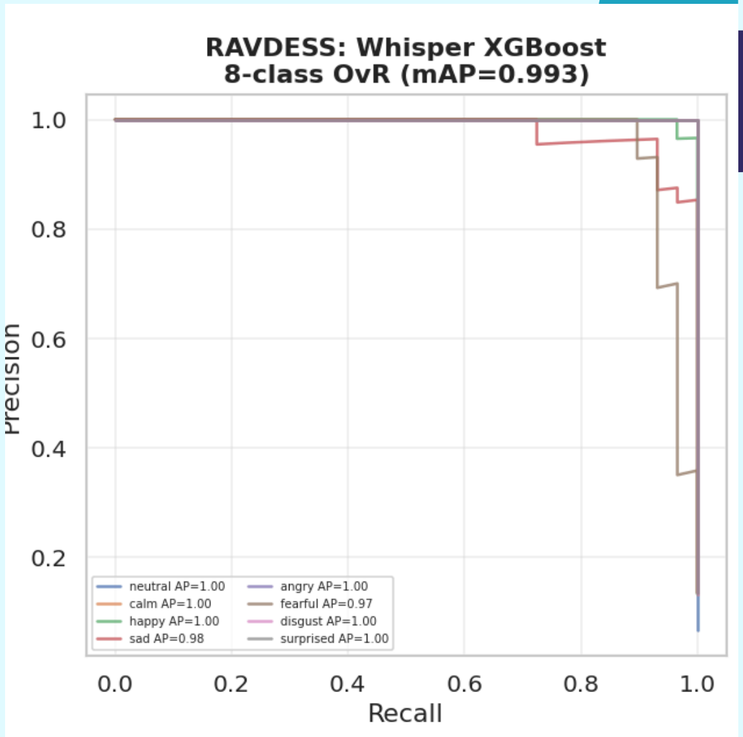
- MFCC: errors across multiple pairs — Calm↔Neutral, Sad↔Fearful, Happy↔Surprised
- Whisper: errors collapse to Calm ↔ Neutral only



- DAIC-WOZ: STRONG FALSE-NEGATIVE BIAS (DEPRESSED PREDICTED AS HEALTHY)

ROC & PR CURVES





Key points:

- DAIC-WOZ: Fusion MLP achieves better precision at high recall than acoustic-only XGB
- Threshold tuning is critical — sensitivity = 0.857 at threshold 0.20 vs. 0.571 at 0.45

QUALITATIVE RESULTS

Successes:

- Angry speech (loud, fast) → Angry · Sad speech (slow, flat) → Sad
- Long pauses + low F0 (PHQ-8=16) → Depressed
- Normal rate + engagement (PHQ-8=3) → Healthy

Failures:

True	Predicted	Reason
Calm	Neutral	Identical low-energy prosody — indistinguishable even for Whisper
Depressed(PHQ-8=12)	Healthy	Recently medicated — acoustic symptoms normalized
Healthy (PHQ-8=6)	Depressed	Introverted speaker — slow rate falsely flagged
Stressed (SWELL)	Relaxed	Mild cognitive stressor below HRV detection threshold

Error Analysis

- Error Patterns & Root Causes

RAVDESS:

- MFCC averages remove temporal emotion cues
- Calm vs Neutral confusion persists

DAIC-WOZ:

- Only 30 positive samples limit performance
- False negatives: masked/medicated patients
- False positives: introverted/non-native speakers

MODMA:

- High-dimensional Whisper embeddings cause noise issues

SWELL:

- Time Pressure vs Interruption → similar HRV patterns

Observations:

- Feature quality > model architecture — Whisper gap (+48.8 pp) dwarfs any tuning gain
- XGBoost beats MLP on every small tabular dataset in this project

LIMITATIONS:

Limitation	Impact
MODMA N=8 test set	Each error = ±0.125 F1 — not statistically reliable
Frozen Whisper encoder	Suboptimal for clinical / non-English speech
Session-level aggregation	Loses within-interview temporal dynamics
SWELL overfitting unresolved	Train 99.8% vs. val 30.7% — not deployable

CONCLUSION

RESULT SUMMARY:

Task	F1	Key Finding
RAVDESS	0.974	Whisper near-ceiling on acted speech
DAIC-WOZ	0.607	Limited by 30 positive training examples
MODMA	0.873	Classical features win when domain mismatches
SWELL	0.508	Naturalistic stress remains unsolved

THANK YOU